

GEORGIOS TRASKAS

AI & ML Engineer | Production Python Backends

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PROFESSIONAL SUMMARY

AI and ML Engineer with 8 years of experience building production-grade data science solutions. For 4+ years at Predictive Fitness, delivered ML-driven predictive models, AWS data workflows, and production observability infrastructure, reporting directly to the Head of Science and Innovation and to the Founder and CEO. Completed 90+ global consulting engagements (Upwork, 100% Job Success Score) across healthcare, energy, and sports science, delivering deep learning models, automated pipelines, and AI-powered applications. Experienced in designing multi-layer security guardrails, vector retrieval architectures, and structured LLM output contracts for production environments. Built and productionized agentic infrastructure: an MCP server taken from Python proof-of-concept to TypeScript production, and autonomous multi-agent systems combining fine-tuned LLMs, RAG, and classic ML.

CORE TECHNICAL SKILLS

RAG & LLM Engineering: Retrieval-Augmented Generation (RAG), Hybrid BM25 + Vector Search, Reciprocal Rank Fusion, Reranking, Prompt Engineering, AI Agents, MCP (Model Context Protocol), Multi-Agent Orchestration, LLM Fine-Tuning (QLoRA)

Frameworks & Orchestration: LangChain, FastAPI (async), Flask, OpenAI API, Google Gemini, Anthropic Claude, Modal (serverless GPU), Gradio

Vector & Relational Databases: pgvector (HNSW indexing), ChromaDB, PostgreSQL, Supabase, DynamoDB, AWS Athena

Machine Learning: XGBoost, Scikit-Learn, TensorFlow, PyTorch, Classification, Regression, Time-Series Forecasting

Cloud & Infrastructure: AWS (Lambda, ECS Fargate, S3, Athena, RDS), Docker, GitHub Actions CI/CD, Render, Cloudflare

Security & Production: Row-Level Security, Sentry (tracing, profiling), Grafana, SlowAPI rate limiting

Languages: Python, TypeScript, SQL, R

Data & MLOps: Feature Engineering, Data Pipelines, ETL, Model Monitoring, Pydantic contracts, Semantic caching

Analytics & Reporting: Statistical modeling, hypothesis testing, cohort analysis, causal inference (do-calculus, Pearl)

PROFESSIONAL EXPERIENCE

Predictive Fitness, Inc. | Data Scientist & ML Engineer | May 2022 – Present (part-time since May 2026) | Remote

- Reported directly to the Head of Science and Innovation and to the Founder and CEO, presenting model performance updates and data-driven recommendations in weekly executive meetings.
- Designed and deployed production ML pipelines processing high-volume athlete fitness telemetry, including time-series forecasting and classification models for workout outcome prediction.
- Built automated AWS data workflows using Lambda, ECS Fargate, and Athena to ingest and process large-scale datasets, replacing manual handling and accelerating modeling cycles.
- Developed science-based physiological models using exercise physiology equations and empirical research.
- Applied causal inference (Pearl's framework, DAGs, do-calculus) to trial conversion prediction helping the marketing team to directly action targeting campaigns and user nudges.
- Integrated live GraphQL APIs with Python backend workflows for real-time feature engineering and data ingestion.
- Proposed and built the company's MCP (Model Context Protocol) server exposing athlete data as tools to any MCP-compliant AI client (Claude, ChatGPT): Python proof-of-concept validated against live AWS data over VPN, then productionized in TypeScript with Streamable HTTP transport, Prisma over GraphQL as the typed data layer, and a RAG layer for training-metric knowledge.
- Surfaced a silent agentic failure mode – tool mis-selection as the toolkit grows – by testing across low-reasoning AI clients and defined the mitigation: unambiguous tool naming plus a golden query set gating every new tool before release.
- Redesigned production alerting architecture using Grafana and Sentry, reducing false alert noise by over 90% from dozens of daily false positives to fewer than five genuine alerts per week, using evidence-gate thresholds and seasonal baseline comparisons.

Data Science Solutions | Independent AI & Data Science Consultant | Jul 2020 – Present | Remote

- Completed 90+ client engagements globally via Upwork with a 100% Job Success Score, delivering predictive models, ML pipelines, and data analytics solutions across healthcare, energy, and retail sectors for clients in the US, Europe, and Israel.
- **MedTech (Israel):** Trained and optimized a deep learning classification model for embedded microcontroller deployment, achieving 97% accuracy with TFLite quantization for hardware-constrained production.

Centre for Research & Technology Hellas | Chemist & Automation Engineer | Dec 2007 – Jul 2022 | Greece

- Automated analytical data pipelines, increasing system throughput by 80% and reducing delivery times by 50%.
- Applied statistical analysis and experimental design to research projects, delivering reporting tools consumed by scientific teams and management.

SELECTED PROJECTS

MolekitChen, AI/RAG Food-Science iOS App | 2026, live on App Store

- Architected and shipped a full-stack AI mobile application grounding food-science and nutrition answers in peer-reviewed research with live citations and evidence grades.
- **RAG pipeline:** hybrid BM25 + vector search via Reciprocal Rank Fusion, reranking, lost-in-the-middle mitigation, approximately 1,600 indexed CC0/CC-BY papers.
- **Five-layer input guardrail stack:** prompt injection defense, language filtering, medical scope blocking, Pydantic structured output contracts with auto-retry, and citation cross-validation.
- Database-level Row-Level Security, deployed on Render with Sentry tracing and SlowAPI rate limiting.
- **Stack:** Python, FastAPI, LangChain, Google Gemini Flash Lite, Supabase (PostgreSQL + pgvector HNSW), Swift/SwiftUI, Docker, GitHub Actions CI/CD.

Wine-VFM, Autonomous Multi-Agent Bargain Hunter | 2026, live demo on Modal

- Built an **agentic** system that hunts wine bargains: an LLM planner runs a tool-calling loop over a scanner pulling live listings from a real online wine shop, three value-for-money estimators, a fitted ensemble, and a push-notification agent, with persistent memory of past finds.
- **Estimators:** fine-tuned Llama 3.2 3B served on Modal GPU, RAG over ~88K embedded tasting notes (ChromaDB) with a frontier LLM, and a neural network; blended with regression-fitted ensemble weights and evaluated with MAE/R².
- **Serverless economics:** CPU web app scales to zero, GPU wakes only during a hunt; artifacts and memory persist on a Modal Volume.
- **Stack:** Python, Modal, PyTorch, ChromaDB, OpenAI API, Scikit-Learn, Gradio. Code: github.com/gtraskas/wine-vfm

EDUCATION

MSc, Chemical Technology | Aristotle University of Thessaloniki | 2004

BSc, Chemistry | Aristotle University of Thessaloniki | 2002

CERTIFICATIONS

IBM Generative AI (2026) • Claude AI, Anthropic (2026) • AutoML (2022) • AWS Cloud Essentials (2022)

ML Engineer Nanodegree, Udacity (2020) • Google Big Data & ML Fundamentals (2019) • TensorFlow Developer (2019) • Stanford ML, Coursera (2018) • Data Analyst, Udacity (2017)